Potential Energy and Energy Conservation

- Work
- Kinetic Energy
- Work-Kinetic Energy Theorem
- Gravitational Potential Energy
- Elastic Potential Energy
- Work-Energy Theorem
- Conservative and Non-conservative Forces
- Conservation of Energy

From the book: Physics for future presidents.
Richard Muller. Highly recommended for Science majors

Table 1.1 Energy per Gram

Object	Calories (or watt-hours)	Joules	Compared to TNT
Bullet (at sound speed, 1000 ft/s)	0.01	40	0.015
Battery (auto)	0.03	125	0.05 x 20
Battery (rechargeable computer)	0.1	400	0.15 x 10
Flywheel (at 1 km/s)	0.125	500	0.2
Battery (alkaline flashlight)	0.15	600	0.23
TNT (the explosive trinitrotoluene)	0.65	2700	1
Modern high explosive (PETN)	1	4200	1.6
Chocolate chip cookies	5	21,000	8
Coal	6	27,000	10
Butter	7	29,000	11
Alcohol (ethanol)	6	27,000	10
Gasoline	10	42,000	15
Natural gas (methane, CH_4)	13	54,000	20
Hydrogen gas or liquid (H_2)	26	110,000	40
Asteroid or meteor (30 km/s)	100	450,000	165
Uranium-235	20 million	82 billion	30 million

Note: Many numbers in this table have been rounded off.

Energy Review

- Kinetic Energy
  - Associated with movement of members of a system
- Potential Energy
  - Determined by the configuration of the system
  - Gravitational and Elastic
- Internal Energy
  - Related to the temperature of the system

Kinetic Energy and Work

- Kinetic energy associated with the motion of an object

$$KE = \frac{1}{2} m v^2$$

- Scalar quantity with the same unit as work
- Work is related to kinetic energy

$$\frac{1}{2} m v^2 - \frac{1}{2} m v_0^{\ 2} = (F_{net} \cos\theta)\Delta x$$

$$= \int_{x_i}^{x_f} \mathbf{F} \cdot d\mathbf{r}$$

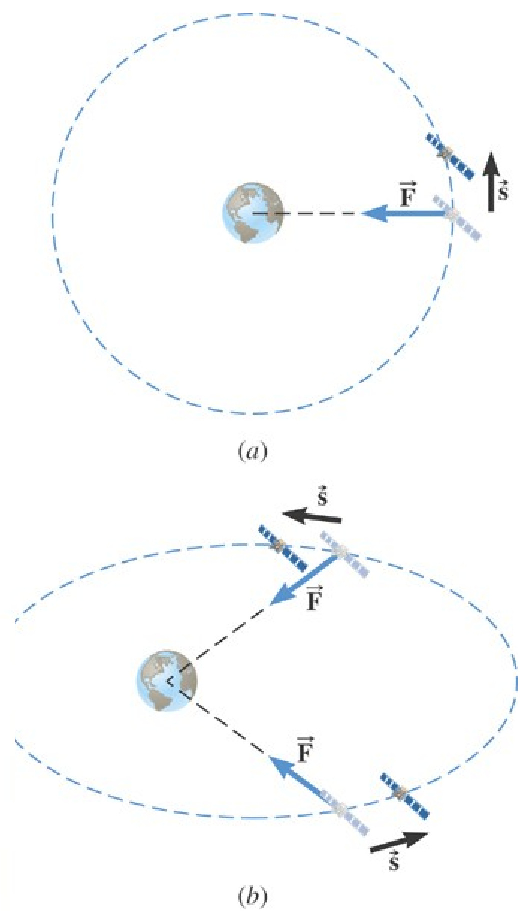
Units: N-m or J

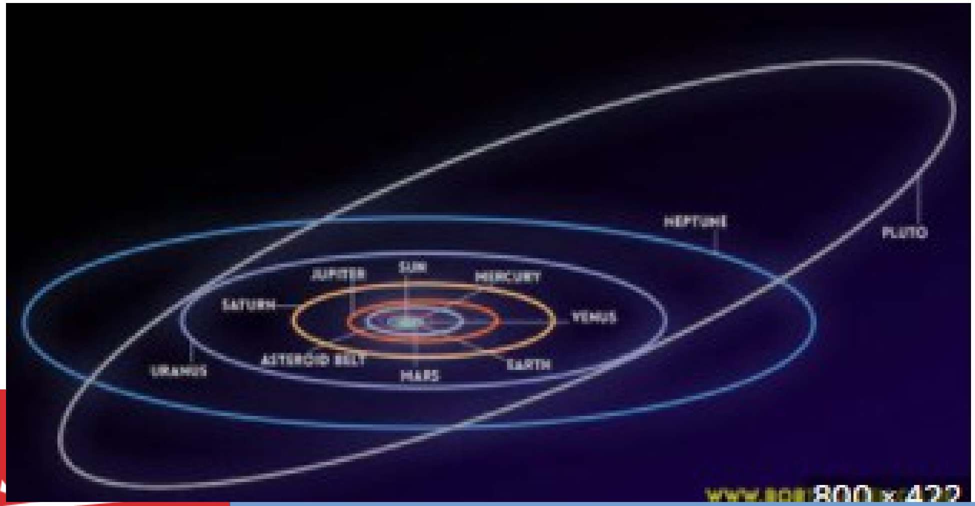
$$W_{net} = KE_f - KE_i = \Delta KE$$

Conceptual Example 6 Work and Kinetic Energy

A satellite is moving about the earth
in a circular orbit and an elliptical orbit.
For these two orbits, determine whether
the kinetic energy of the satellite
changes during the motion. Why?

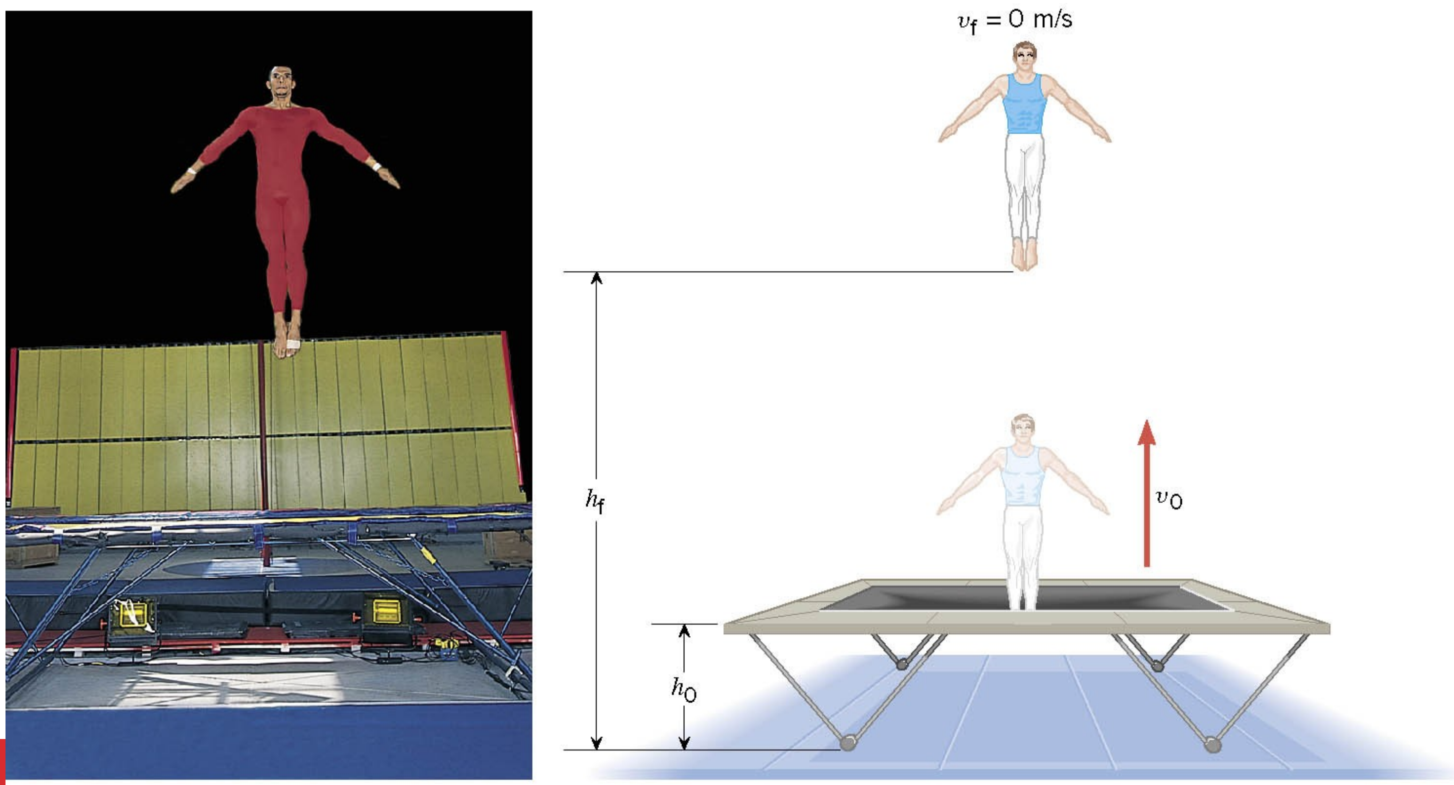
Hint: work is done if there is a component
Along the path





Example 7 A Gymnast on a Trampoline

The gymnast leaves the trampoline at an initial height of 1.20 m and reaches a maximum height of 4.80 m before falling back down. What was the initial speed of the gymnast?



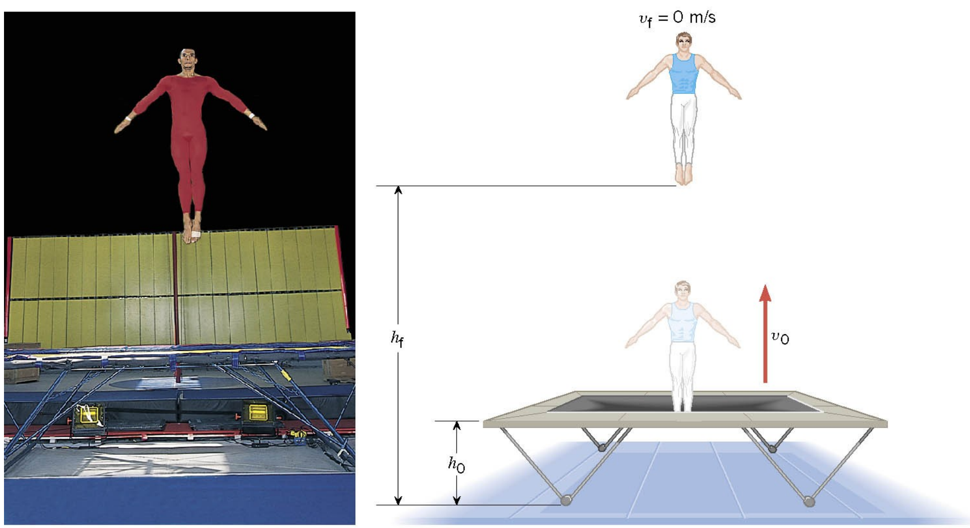
(a) (b)

$$W = \tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_o^2$$

$$W_{\text{gravity}} = mg\left(h_o - h_f\right)$$

$$mg\left(h_o - h_f\right) = -\tfrac{1}{2}mv_o^2$$

$$v_o = \sqrt{-2g\left(h_o - h_f\right)}$$



(a) (b)

$$v_o = \sqrt{-2\left(9.80\, m/s^2\right)\left(1.20\ \text{m} - 4.80\ \text{m}\right)} = 8.40\, m/s$$

Conservation of Mechanical Energy

- By the work-energy theorem, the change in an object's kinetic energy equals the net work done *on* the object:

$$\Delta K = W_{net}$$

- When only conservative forces act, the net work is the negative of the potential-energy change: $W_{net} = -\Delta U$
- Therefore when only conservative forces act, any change in potential energy is compensated by an opposite change in kinetic energy:

$$\Delta K + \Delta U = 0$$

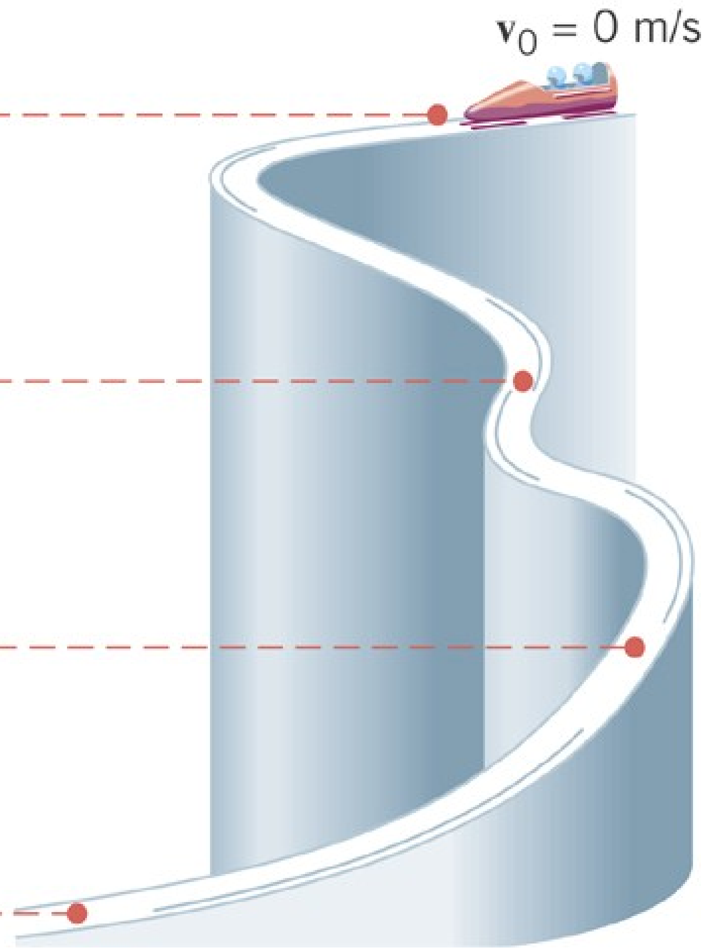
- Equivalently,

$$K + U = \text{constant} = K_0 + U_0$$

- Both these equations are statements of the law of **conservation of mechanical energy.**

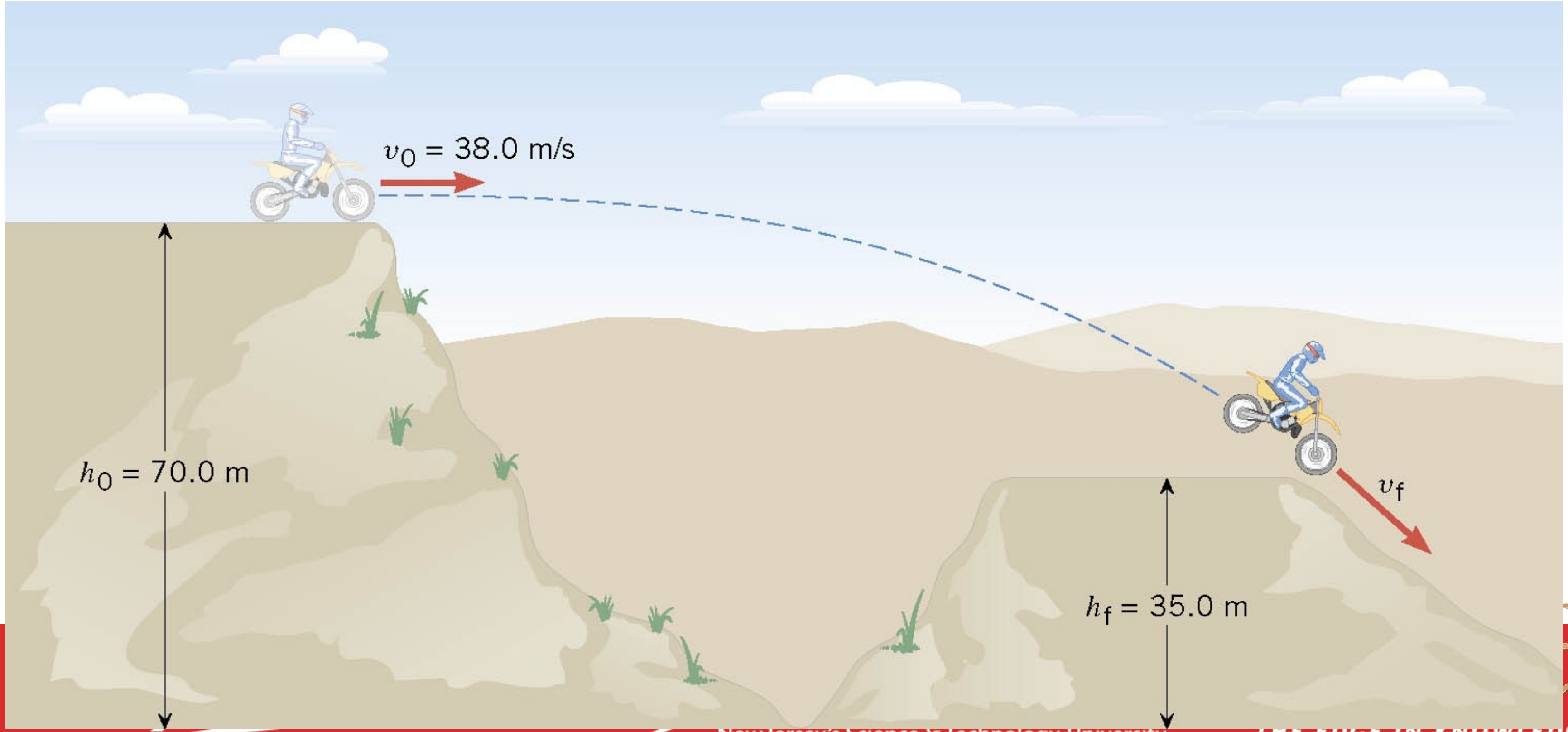
6.5 The Conservation of Mechanical Energy

KE	PE	E = KE + PE
0 J	600 000 J	600 000 J
200 000 J	400 000 J	600 000 J
400 000 J	200 000 J	600 000 J
600 000 J	0 J	600 000 J

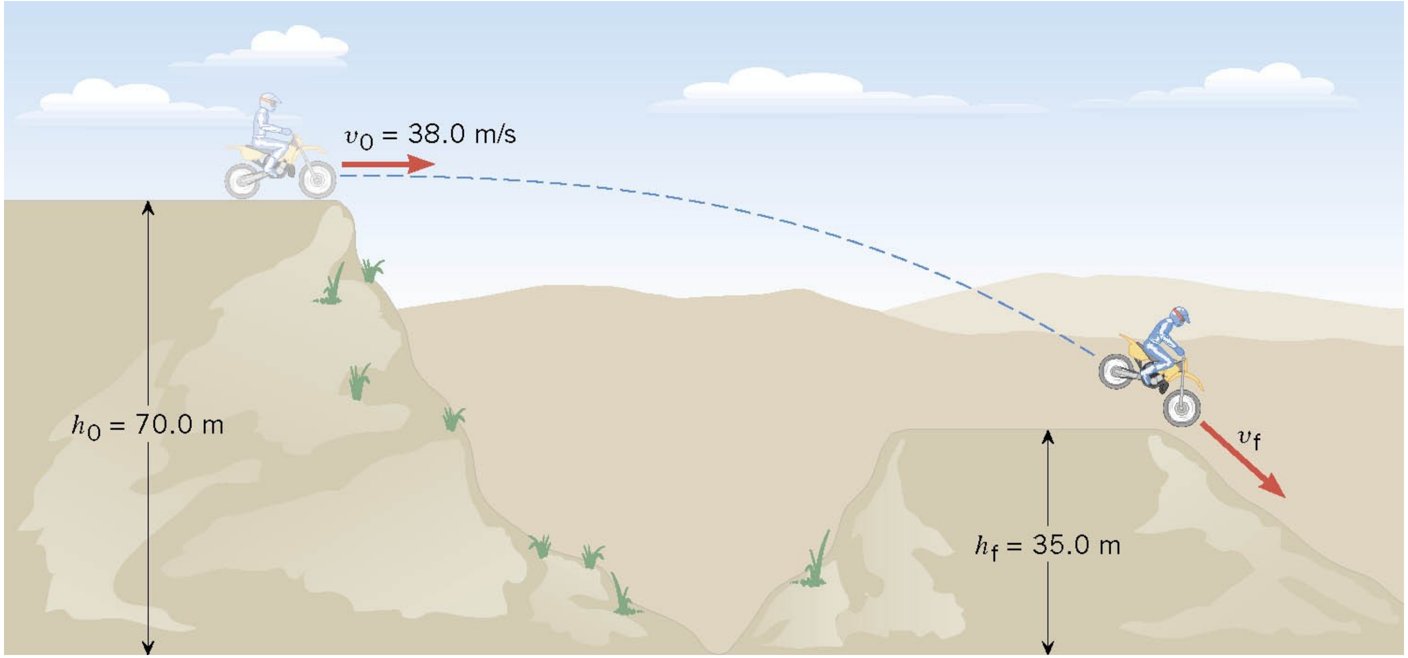


Example 8 A Daredevil Motorcyclist

A motorcyclist is trying to leap across the canyon by driving horizontally off a cliff 38.0 m/s. Ignoring air resistance, find the speed with which the cycle strikes the ground on the other side.



6.5 The Conservation of Mechanical Energy

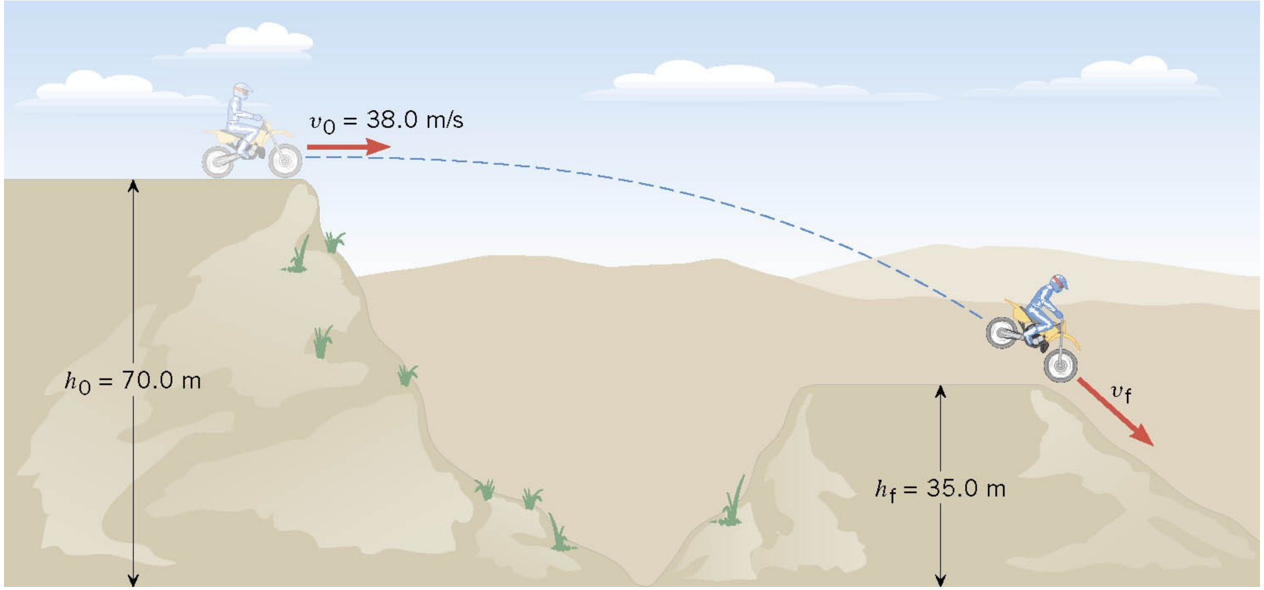


$$E_f = E_o$$

$$mgh_f + \tfrac{1}{2}mv_f^2 = mgh_o + \tfrac{1}{2}mv_o^2$$

$$gh_f + \tfrac{1}{2}v_f^2 = gh_o + \tfrac{1}{2}v_o^2$$

6.5 The Conservation of Mechanical Energy



$$gh_f + \tfrac{1}{2}v_f^2 = gh_o + \tfrac{1}{2}v_o^2$$

$$v_f = \sqrt{2g\left(h_o - h_f\right) + v_o^2}$$

$$v_f = \sqrt{2\left(9.8\,\mathrm{m/s^2}\right)\left(35.0\mathrm{m}\right) + \left(38.0\,\mathrm{m/s}\right)^2} = 46.2\,\mathrm{m/s}$$

Conservation of Energy

- ***Energy is conserved***
  - This means that energy cannot be created nor destroyed
  - If the total amount of energy in a system changes, it can only be due to the fact that energy has crossed the boundary of the system by some method of energy transfer





Ways to Transfer Energy Into or Out of A System

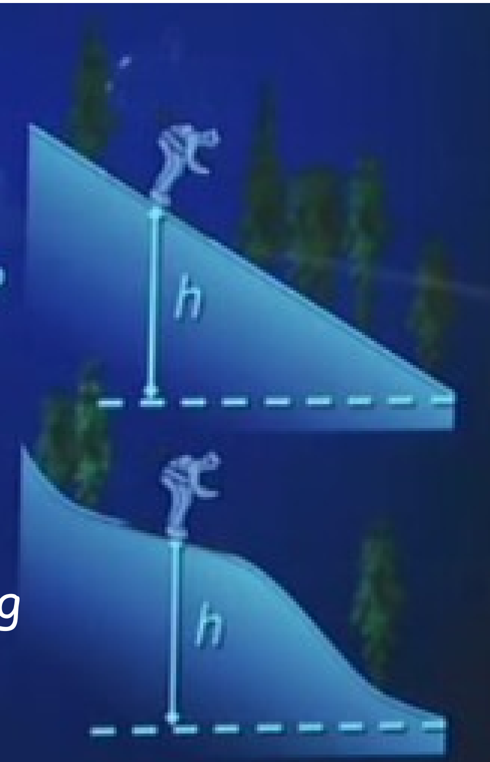
- ***Work*** – transfers by applying a force and causing a displacement of the point of application of the force
- ***Mechanical Waves*** – allow a disturbance to propagate through a medium
- ***Heat*** – is driven by a temperature difference between two regions in space
- ***Matter Transfer*** – matter physically crosses the boundary of the system, carrying energy with it
- ***Electrical Transmission*** – transfer is by electric current
- ***Electromagnetic Radiation*** – energy is transferred by electromagnetic waves

Problem-Solving Strategy

- Define the system
- Select the location of zero gravitational potential energy
  - Do *not* change this location while solving the problem
- Identify two points the object of interest moves between
  - One point should be where information is given
  - The other point should be where you want to find out something

Energy Conservation: Two Skiers

Start from rest and ski down frictionless slopes, vertical drop h. How fast are they going at the bottom?



Using Newton's second law would be challenging for case#2. The acceleration is Not constant. So calculus is complex here.

If you use the conservation of energy, its easy!

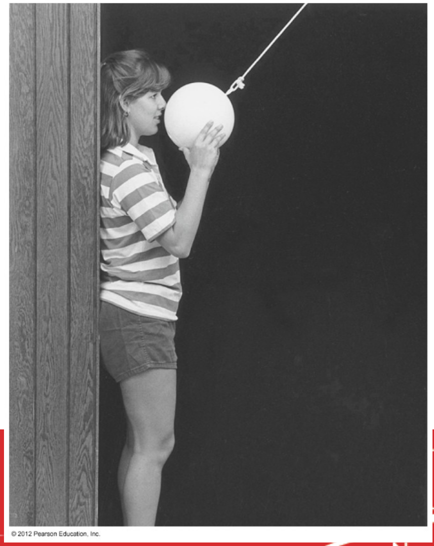
Clicker Question 3

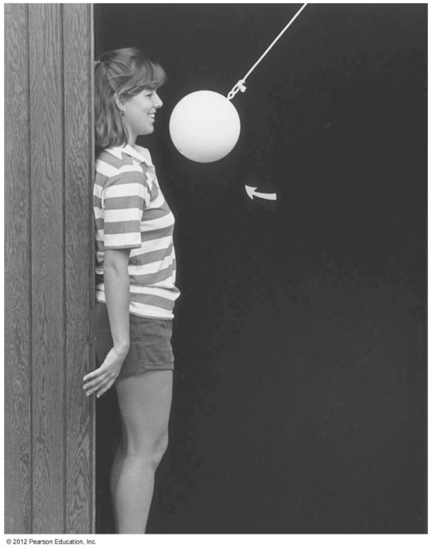
- A bowling ball is tied to the end of a long rope and suspended from the ceiling. A student stands at one side of the room, holds the ball to her nose, and then releases it from rest. Should she duck as it swings back?

See video Walter lewin

A. She should duck!

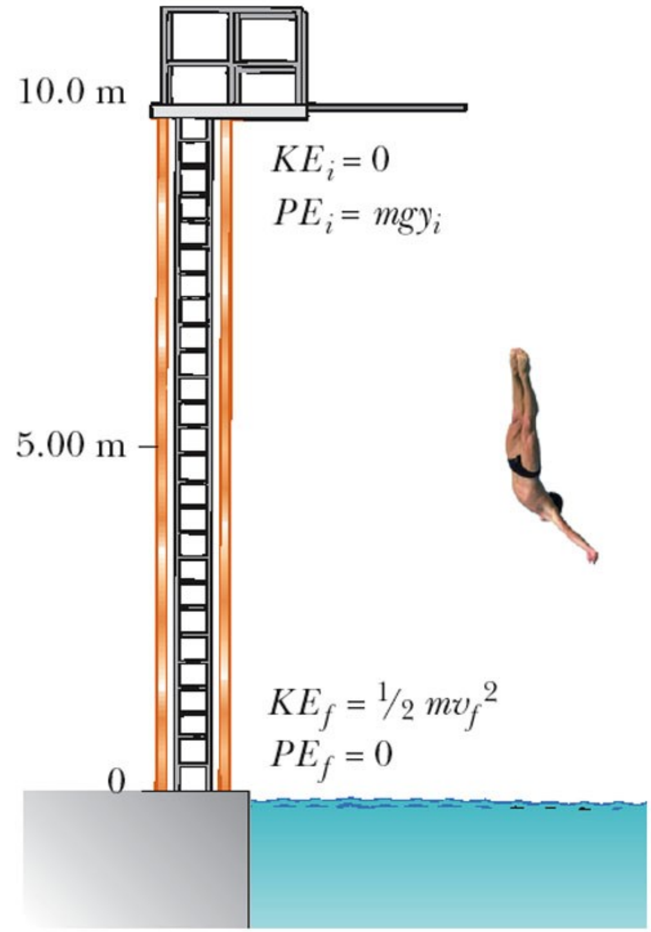
B. She does not need to duck.





Platform Diver

- A diver of mass m drops from a board 10.0 m above the water's surface. Neglect air resistance.
- (a) Find is speed 5.0 m above the water surface
- (b) Find his speed as he hits the water



Platform Diver

- (a) Find his speed 5.0 m above the water surface

$$\frac{1}{2}mv_i^2+mgy_i=\frac{1}{2}mv_f^2+mgy_f$$

$$0+gy_i=\frac{1}{2}v_f^2+mgy_f$$

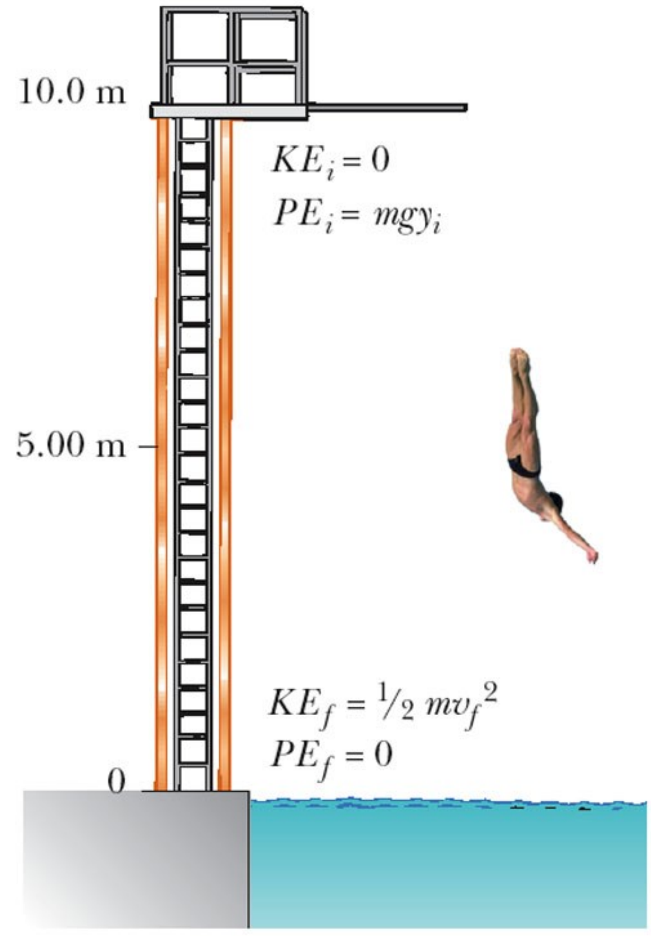
$$v_f=\sqrt{2g\left(y_i-y_f\right)}$$

$$\text{¿}\sqrt{2\left(9.8m/s^2\right)\left(10m-5m\right)}=9.9m/s$$

- (b) Find his speed as he hits the water

$$0+mgy_i=\frac{1}{2}mv_f^2+0$$

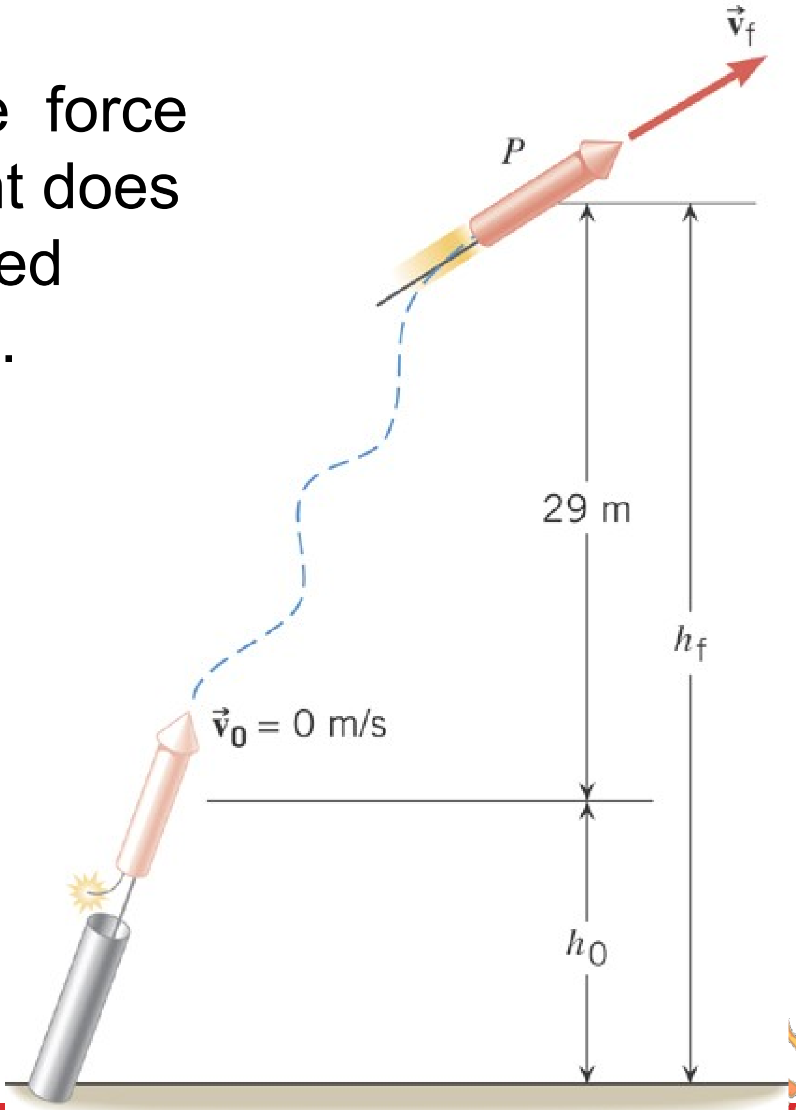
$$v_f=\sqrt{2gy_i}=14m/s$$



Example 11 Fireworks

Assuming that the nonconservative force generated by the burning propellant does 425 J of work, what is the final speed of the rocket. Ignore air resistance.
m=200g

$$W_{nc} = \left(mgh_f + \tfrac{1}{2}mv_f^2\right) - \left(mgh_o + \tfrac{1}{2}mv_o^2\right)$$



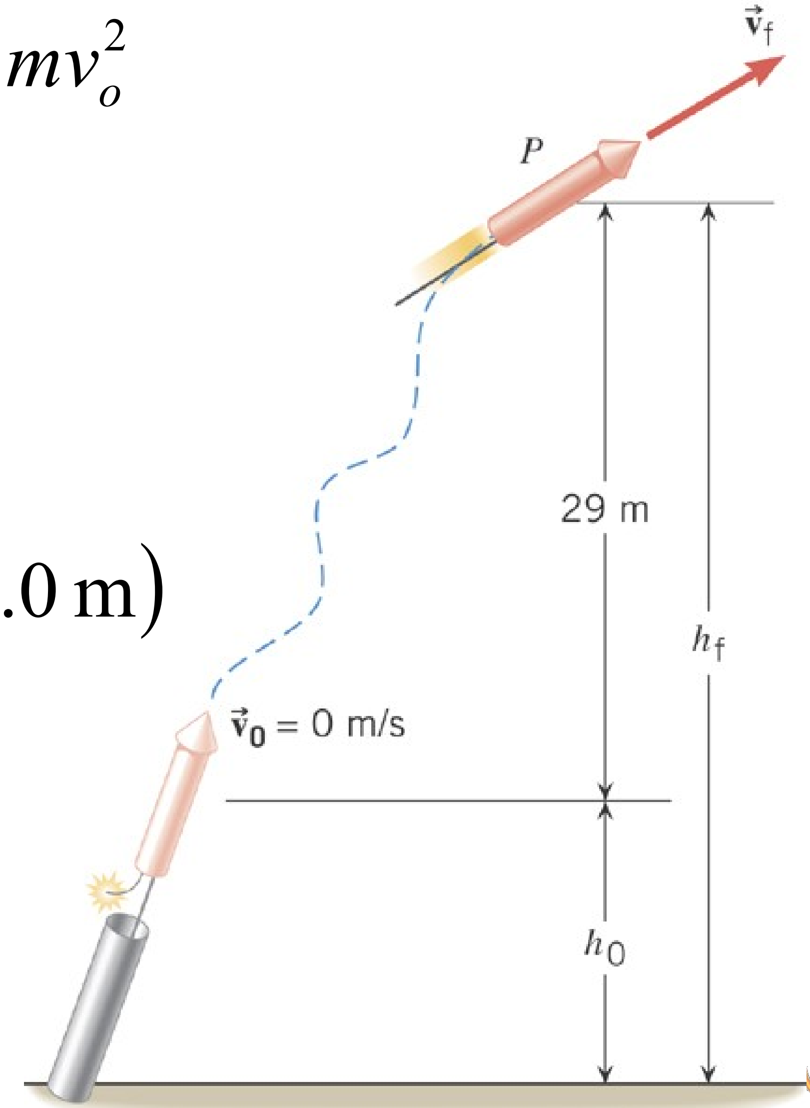
6.6 Nonconservative Forces and the Work-Energy Theorem

$$W_{nc} = mgh_f - mgh_o + \tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_o^2$$

$$W_{nc} = mg\left(h_f - h_o\right) + \tfrac{1}{2}mv_f^2$$

$$425\,\text{J} = (0.20\,\text{kg})\left(9.80\,\text{m/s}^2\right)(29.0\,\text{m}) + \tfrac{1}{2}(0.20\,\text{kg})v_f^2$$

$$v_f = 61\,\text{m/s}$$



Extended Work-Energy Theorem

- The work-energy theorem can be extended to include potential energy:

$$W_{net} = KE_f - KE_i = \Delta KE$$

$$W_{gravity} = PE_i - PE_f \qquad W_s = PE_{si} - PE_{sf}$$

- If we include gravitational force and spring force, then

$$W_{net} = W_{gravity} + W_s$$

$$\left(KE_f - KE_i\right) + \left(PE_f - PE_i\right) + \left(PE_{sf} - PE_{si}\right) = 0$$

$$KE_f + PE_f + PE_{sf} = PE_i + KE_i + KE_{si}$$

Extended Work-Energy Theorem

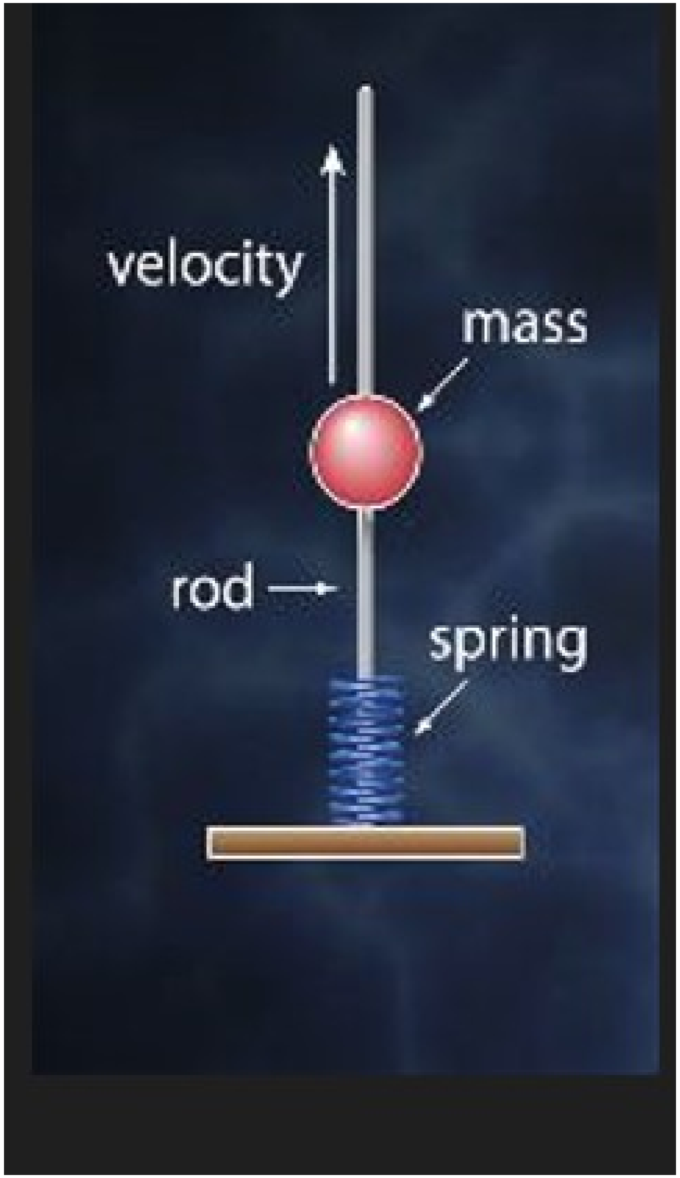
- We denote the total mechanical energy by

$$E = KE + PE + PE_s$$

- Since $$\left(KE + PE + PE_s \right)_f = \left(KE + PE + PE_s \right)_i$$

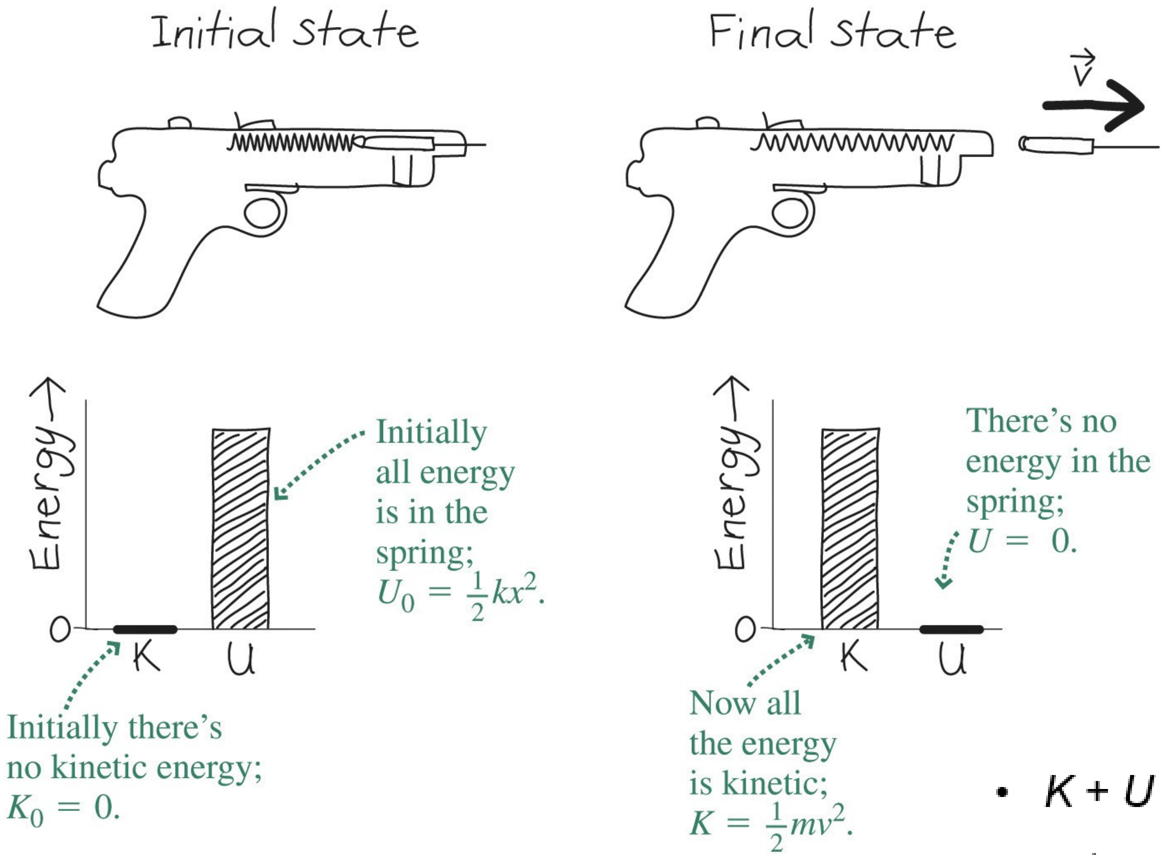
- The total mechanical energy is conserved and remains the same at all times

$$\frac{1}{2}mv_i^2 + mgy_i + \frac{1}{2}kx_i^2 = \frac{1}{2}mv_f^2 + mgy_f + \frac{1}{2}kx_f^2$$



PEg → KE1 + heat → Peel+heat → KE2+heat => Peg

Video. stairs/trampoline



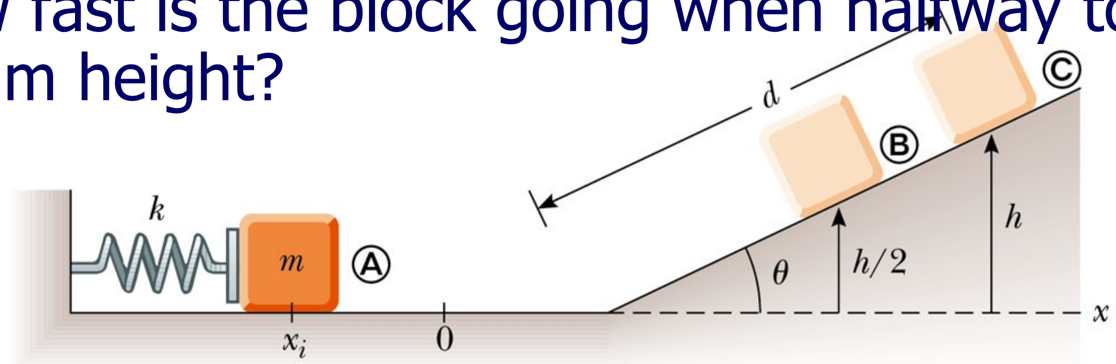
- $K + U = K_0 + U_0$ becomes

  $$\tfrac{1}{2}mv^2 + 0 = 0 + \tfrac{1}{2}kx^2$$

- So $v = \sqrt{k/m}\,x$
  where x is the initial spring compression.

A block projected up a incline

- A 0.5-kg block rests on a horizontal, frictionless surface. The block is pressed back against a spring having a constant of k = 625 N/m, compressing the spring by 10.0 cm to point A. Then the block is released.
- (a) Find the maximum distance d the block travels up the frictionless incline if θ = 30°.
- (b) How fast is the block going when halfway to its maximum height?



A block projected up a incline

- Point A (initial state): $v_i = 0, y_i = 0, x_i = -10\,cm = -0.1\,m$
- Point B (final state): $v_f = 0, y_f = h = d\sin\theta, x_f = 0$

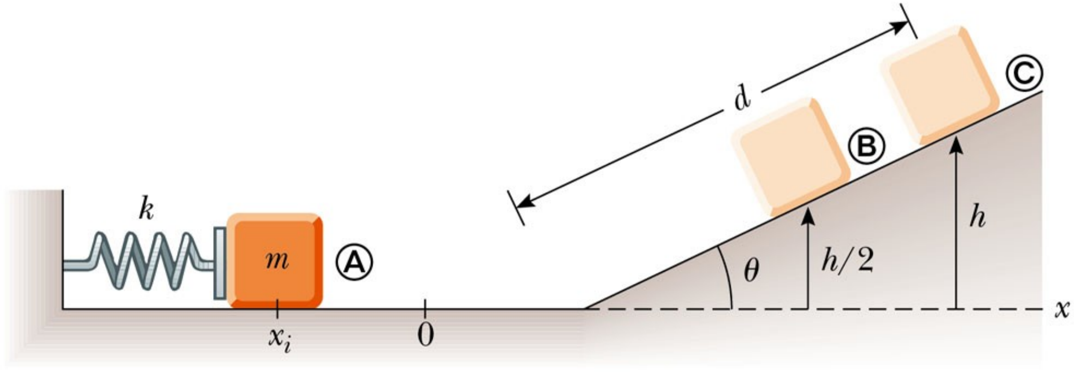
$$\frac{1}{2}mv_i^2 + mgy_i + \frac{1}{2}kx_i^2 = \frac{1}{2}mv_f^2 + mgy_f + \frac{1}{2}kx_f^2$$

$$\frac{1}{2}kx_i^2 = mgy_f = mgd\sin\theta$$

$$d = \frac{\frac{1}{2}kx_i^2}{mg\sin\theta}$$

$$\text{¿}\,\frac{0.5(625\,N/m)(-0.1\,m)^2}{(0.5\,kg)(9.8\,m/s^2)\sin 30^\circ}$$

$$\text{¿}\,1.28\,m$$



A block projected up a incline

- Point A (initial state): $v_i=0\,, y_i=0\,, x_i=-10\,cm=-0.1\,m$
- Point B (final state): $v_f=?, y_f=h/2=d\sin\theta/2\,, x_f=0$

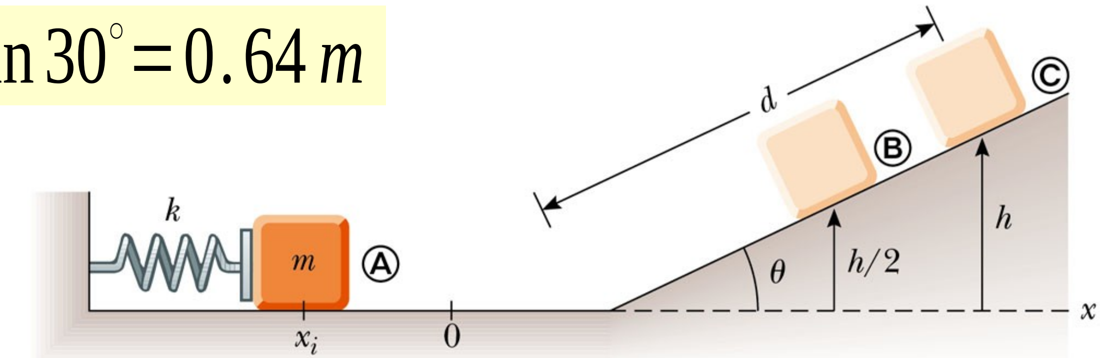
$$\frac{1}{2}mv_i^2+mgy_i+\frac{1}{2}kx_i^2=\frac{1}{2}mv_f^2+mgy_f+\frac{1}{2}kx_f^2$$

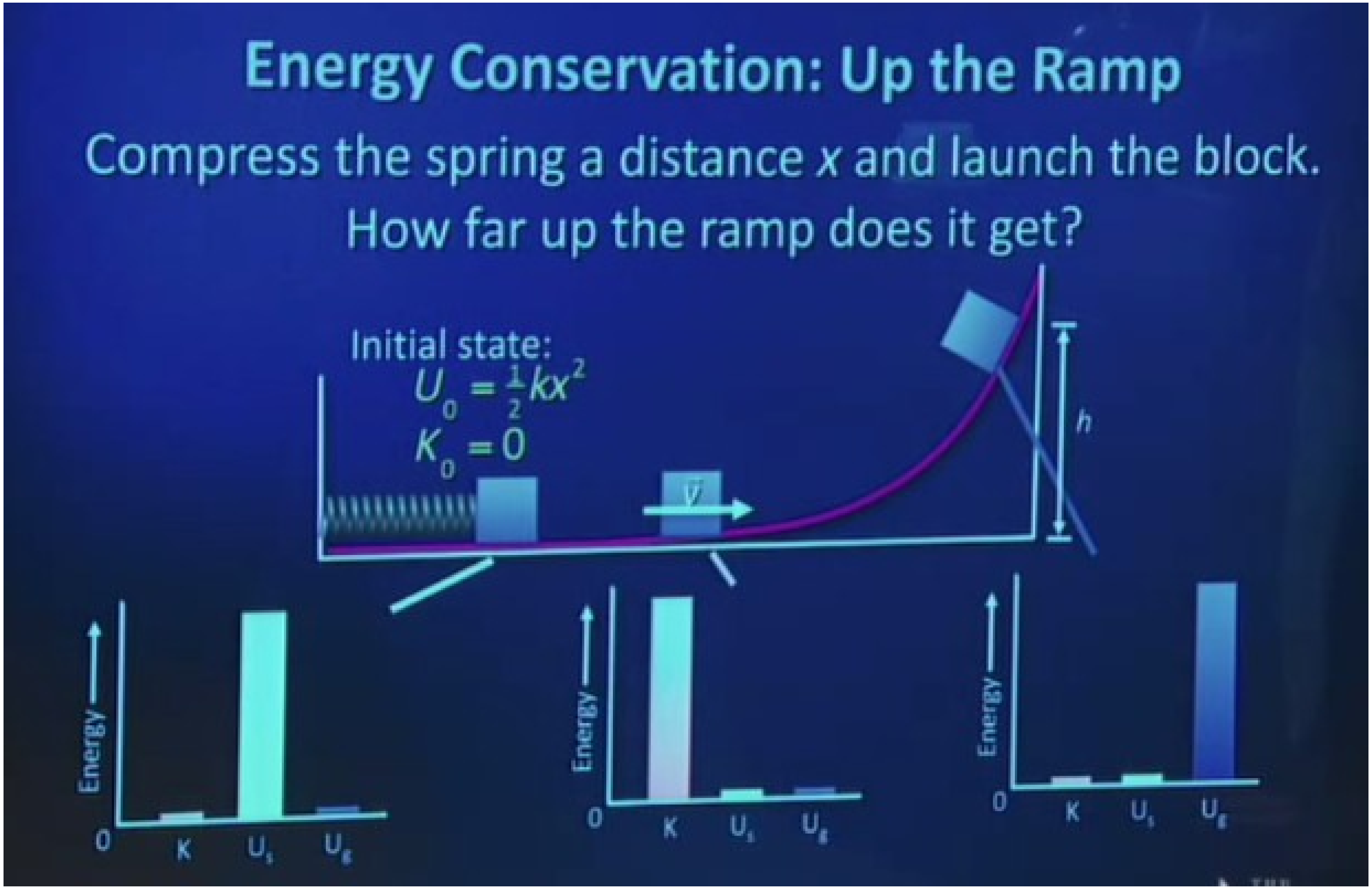
$$\frac{1}{2}kx_i^2=\frac{1}{2}mv_f^2+mg\left(\frac{h}{2}\right) \qquad \frac{k}{m}x_i^2=v_f^2+gh$$

$$h=d\sin\theta=(1.28\,m)\sin 30^\circ=0.64\,m$$

$$v_f=\sqrt{\frac{k}{m}x_i^2-gh/2}$$

$$¿......=2.5\,m/s$$



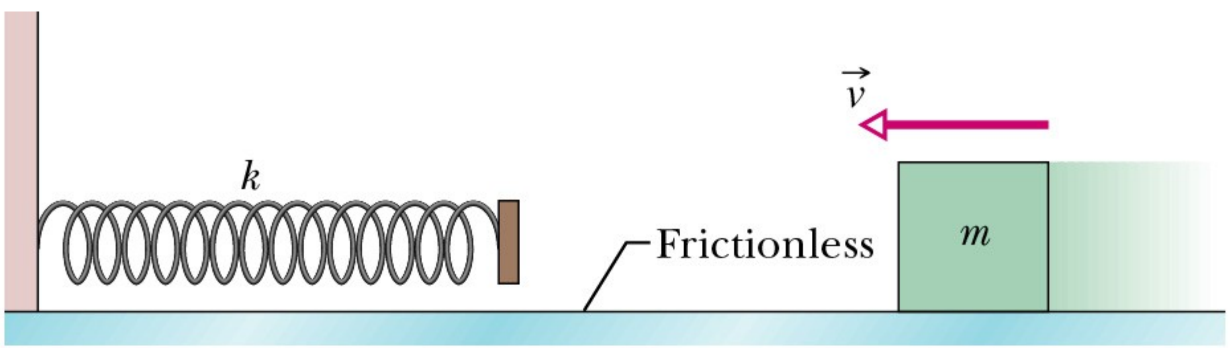

Energy Conservation: Up the Ramp
Compress the spring a distance x and launch the block.
How far up the ramp does it get?
Initial state:
$U_0 = \frac{1}{2}kx^2$
$K_0 = 0$
$\vec{v}$
h
Energy
0
K
U_s
U_g

- just write the equations

Conservation of Mechanical Energy

- A block of mass m = 0.40 kg slides across a horizontal frictionless counter with a speed of v = 0.50 m/s. It runs into and compresses a spring of spring constant k = 750 N/m. When the block is momentarily stopped by the spring, by what distance d is the spring compressed?

- Now there is kinetic friction uk=0.1 . Whats the new distance d.



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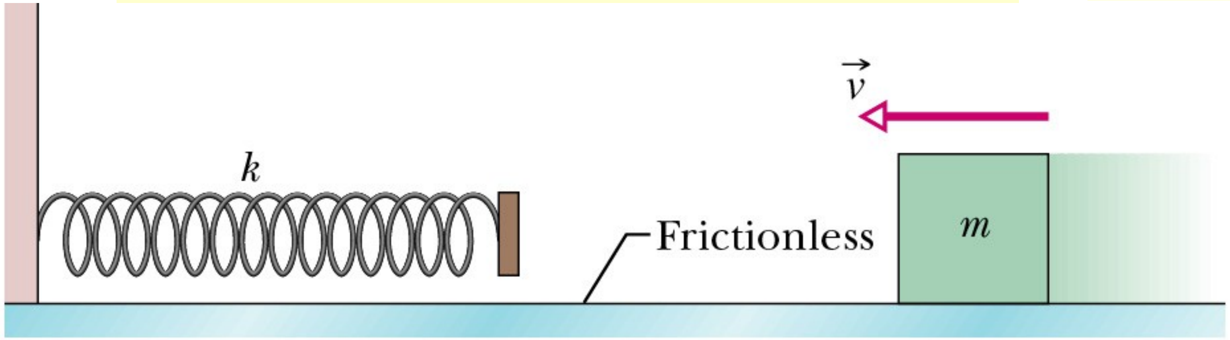
$$W_{nc} = (KE_f + PE_f) - (KE_i + PE_i)$$

$$\frac{1}{2}mv_f^2 + mgy_f + \frac{1}{2}kx_f^2 = \frac{1}{2}mv_i^2 + mgy_i + \frac{1}{2}kx_i^2$$

$$0 + 0 + \frac{1}{2}kd^2 = \frac{1}{2}mv^2 + 0 + 0$$



$$d = \sqrt{\frac{m}{k}v^2} = 1.15\,cm$$

Find an equation for the minimum starting height h that will permit the car to get around the loop without falling off the track

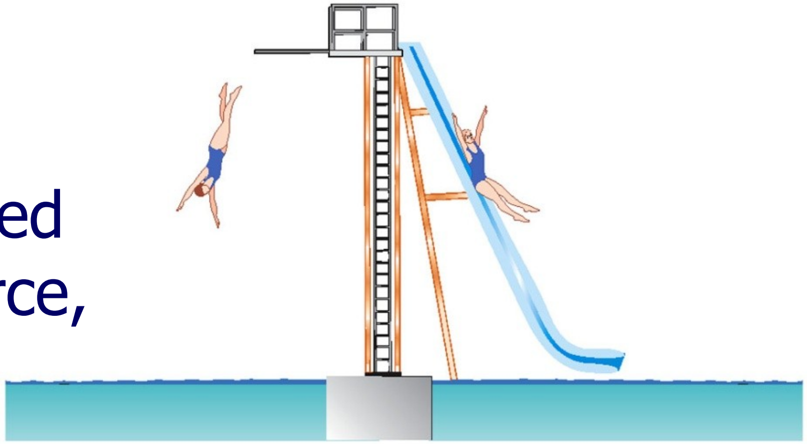
H=5/2 R
Show that.
(top of loop N=0)

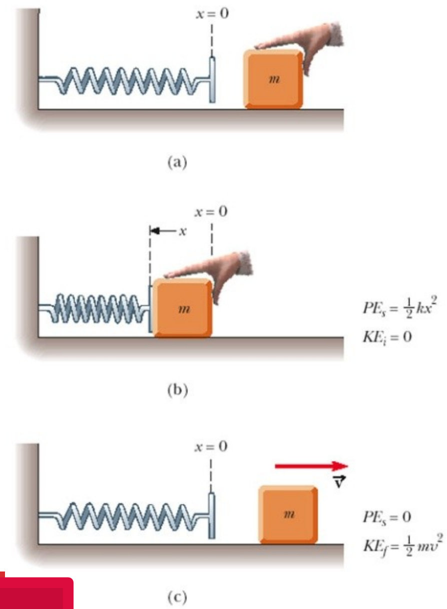
Loop the loop demo or video

https://www.youtube.com/watch?v=dA_UO86MjLY

Types of Forces

- Conservative forces
  - Work and energy associated with the force can be recovered
  - Examples: Gravity, Spring Force, EM forces
- Nonconservative forces
  - The forces are generally dissipative and work done against it cannot easily be recovered
  - Examples: Kinetic friction, air drag forces, normal forces, tension forces, applied forces …





Non conservative Forces and the Work-Energy Theorem

If other forces than gravity are involved (thrust from engine or friction) then write:

PE1 + KE1 + (work done by thrust or you) = PE2 + KE2

Energy is added to the whole system. Ex If you push a cart

ex. :mgh1 + 0.5 m v1^2 *+ push x displacement = mgh2 + 0.5 m V2*^{2}

This is the same as : net work = change in KE

PE1 + KE1 + (work done by friction) = PE2 + KE2

Work done by friction is negative = - friction x displacement

Energy is removed from the whole system. Ex: friction

mgh1 + 0.5 m v1^2 *- friction x displacement = mgh2 + 0.5 m V2*^{2}

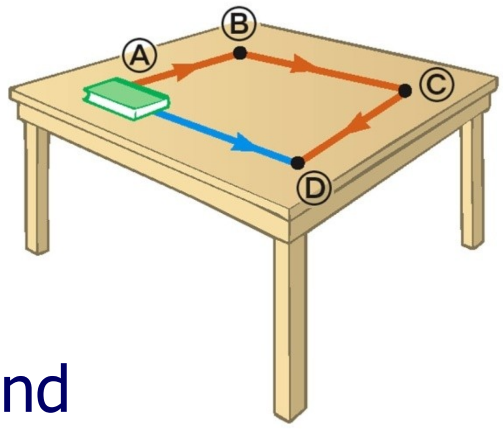
Friction= µk Normal.µk is the coefficient of kinetic friction

Can be work done by air resistance. The work is also negative.

Nonconservative Forces

- A force is nonconservative if the work it does on an object depends on the path taken by the object between its final and starting points.
  - The work depends upon the movement path
  - For a non-conservative force, potential energy can NOT be defined
  - Work done by a nonconservative force

$$W_{nc} = \sum \vec{F} \cdot \vec{d} = -f_k d + \sum W_{otherforces}$$



  - It is generally dissipative. The dispersal of energy takes the form of heat or sound

6.4 Conservative Versus Nonconservative Forces

An example of a nonconservative force is the kinetic frictional force.

$$W = \left(F\cos\theta\right)s = f_k \cos 180^\circ s = -f_k s$$

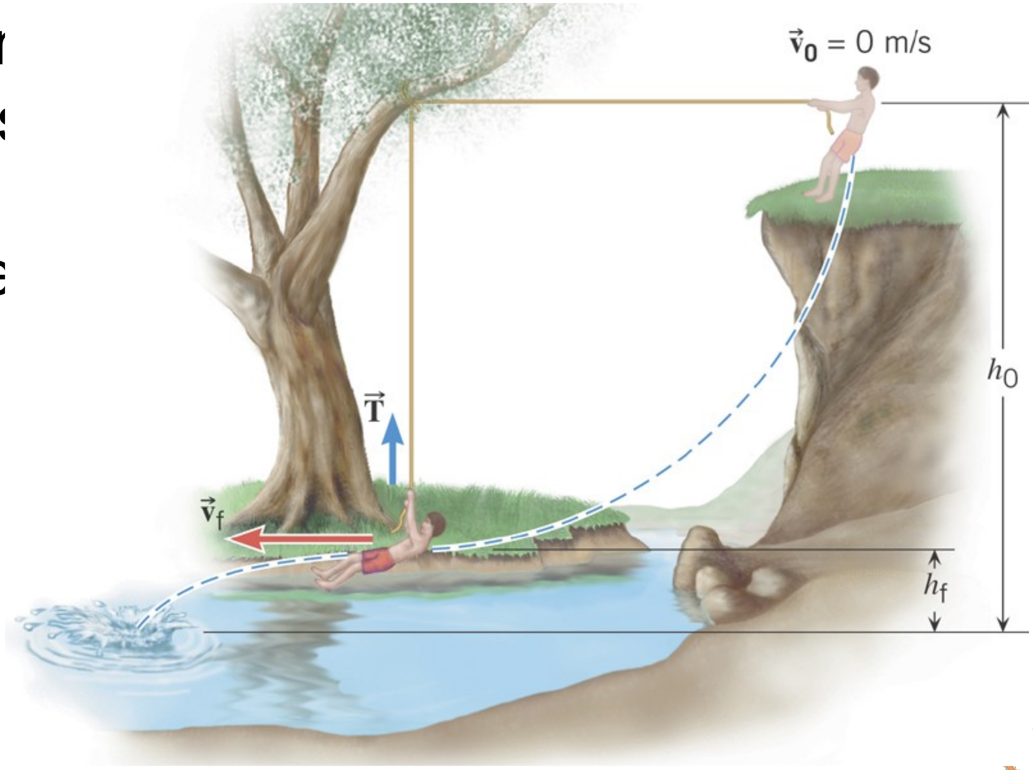
The work done by the kinetic frictional force is always negative. Thus, it is impossible for the work it does on an object that moves around a closed path to be zero.

The concept of potential energy is not defined for a nonconservative force.

Conceptual Example 9 The Favorite Swimming Hole

The person starts from rest, with the rope held in the horizontal position, swings downward, and then lets go of the rope. Three forces act on him: his weight, the tension in the rope, and the force of air resistance.

Can the principle of conservation of energy be used to calculate his final speed?



Problem-Solving Strategy

- Define the system to see if it includes non-conservative forces (especially friction, drag force …)
- Without non-conservative forces

$$\frac{1}{2}mv_f^2+mgy_f+\frac{1}{2}kx_f^2=\frac{1}{2}mv_i^2+mgy_i+\frac{1}{2}kx_i^2$$

- With non-conservative forces

$$W_{nc}=\left(KE_f+PE_f\right)-\left(KE_i+PE_i\right)$$

$$-fd+\sum W_{otherforces}=\left(\frac{1}{2}mv_f^2+mgy_f+\frac{1}{2}kx_f^2\right)-\left(\frac{1}{2}mv_i^2+mgy_i+\frac{1}{2}kx_i^2\right)$$

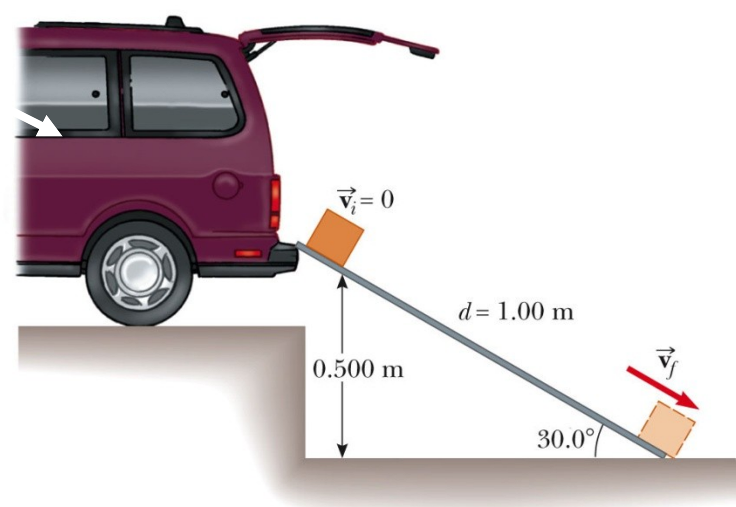
- Select the location of zero potential energy
  - Do *not* change this location while solving the problem
- Identify two points the object of interest moves between
  - One point should be where information is given
  - The other point should be where you want to find out something

Changes in Mechanical Energy for conservative forces

- A 3-kg crate slides down a ramp. The ramp is 1 m in length and inclined at an angle of 30° as shown. The crate starts from rest at the top. The surface friction can be negligible. Use energy methods to determine the speed of the crate at the bottom of the ramp.

$$mv_i^2+mgy_i+\frac{1}{2}kx_i^2)$$

Easy . As usual use:
PE = KE
So V=sqrt(2gh)



Changes in Mechanical Energy for conservative forces

- A 3-kg crate slides down a ramp. The ramp is 1 m in length and inclined at an angle of 30° as shown. The crate starts from rest at the top. The surface friction can be negligible. Use energy methods to determine the speed of the crate at the bottom of the ramp.

$$-fd+\sum W_{otherforces}=(\frac{1}{2}mv_f^2+mgy_f+\frac{1}{2}kx_f^2)-(\frac{1}{2}mv_i^2+mgy_i+\frac{1}{2}kx_i^2)$$

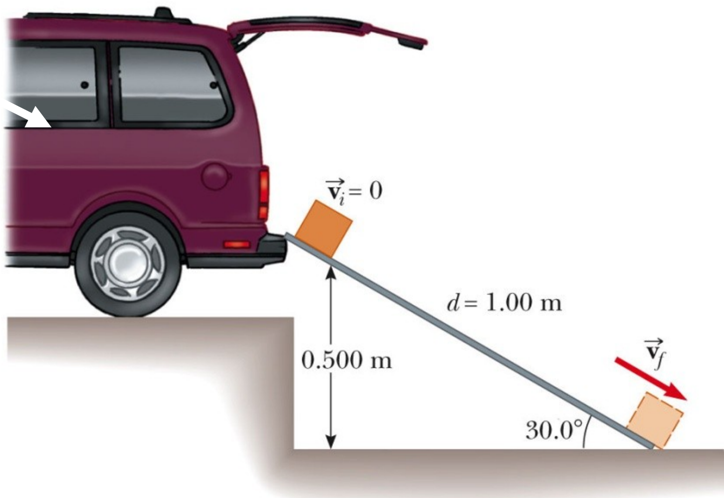
$$(\frac{1}{2}mv_f^2+mgy_f+\frac{1}{2}kx_f^2)=(\frac{1}{2}mv_i^2+mgy_i+\frac{1}{2}kx_i^2)$$

$$d=1m, y_i=d\sin 30^\circ=0.5m, v_i=0$$

$$y_f=0, v_f=?$$

$$(\frac{1}{2}mv_f^2+0+0)=(0+mgy_i+0)$$

$$v_f=\sqrt{2gy_i}=3.1m/s$$

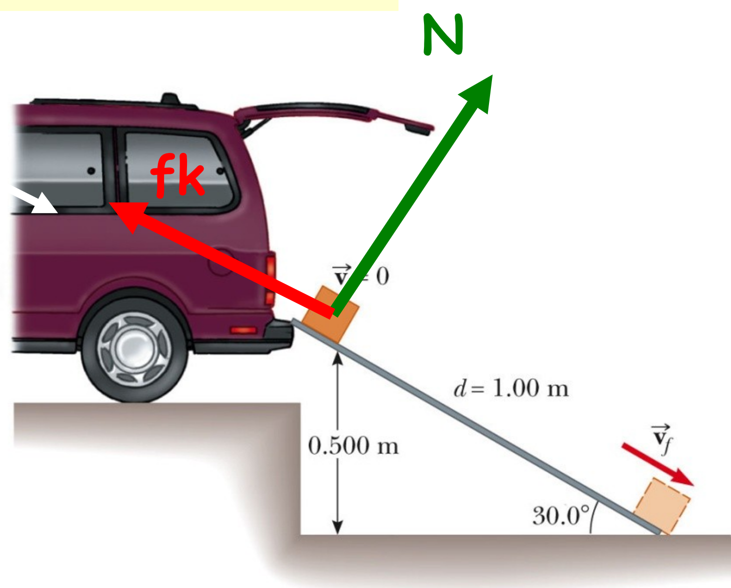


Changes in Mechanical Energy for Non-conservative forces

- A 3-kg crate slides down a ramp. The ramp is 1 m in length and inclined at an angle of 30° as shown. The crate starts from rest at the top. The surface in contact have a coefficient of kinetic friction of 0.15. Use energy methods to determine the speed of the crate at the bottom of the ramp.

PE1 - Wfriction= PE2+ KE2

$$\left(\frac{1}{2}mv_i^2 + mgy_i + \frac{1}{2}kx_i^2\right)$$



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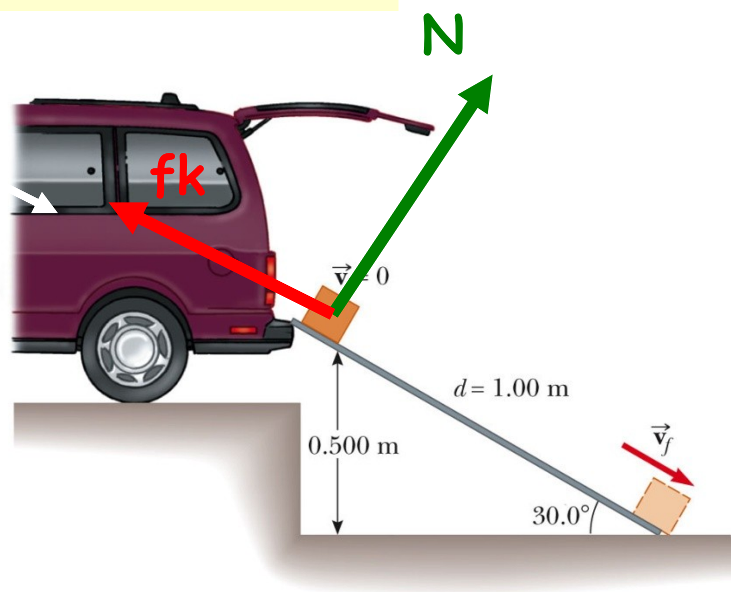
$$-\mu_k Nd+0=\left(\frac{1}{2}mv_f^2+0+0\right)-\left(0+mgy_i+0\right)$$

$$\mu_k=0.15,\, d=1\,m,\, y_i=d\sin 30^\circ=0.5\,m,\, N=?$$

$$N-mg\cos\theta=0$$

$$-\mu_k dmg\cos\theta=\frac{1}{2}mv_f^2-mgy_i$$

$$v_f=\sqrt{2g\left(y_i-\mu_k d\cos\theta\right)}=2.7\,m/s$$



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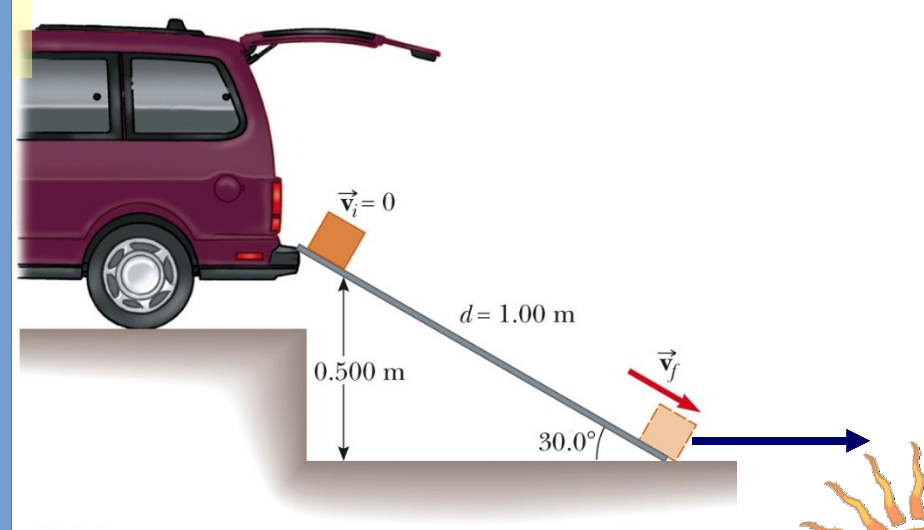
- A 3-kg crate slides down a ramp. The ramp is 1 m in length and inclined at an angle of 30° as shown. The crate starts from rest at the top. The surface in contact have a coefficient of kinetic friction of 0.15. How far does the crate slide on the horizontal floor if it continues to experience a friction force.

All the energy is gone

$$\ldots x_f^2) - (\frac{1}{2}mv_i^2 + mgy_i + \frac{1}{2}kx_i^2)$$



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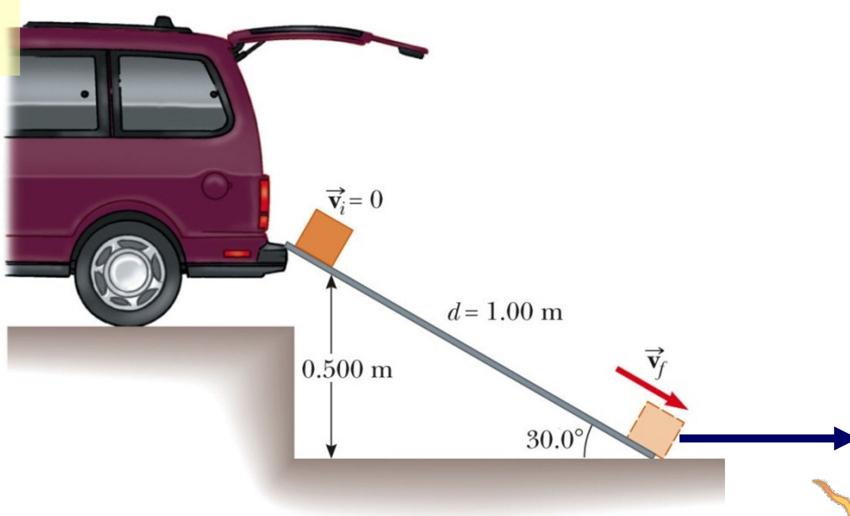
$$-\mu_k Nx+0=(0+0+0)-\left(\frac{1}{2}mv_i^2+0+0\right)$$

$$\mu_k=0.15,\ v_i=2.7\,m/s,\ N=?$$

$$N-mg=0$$

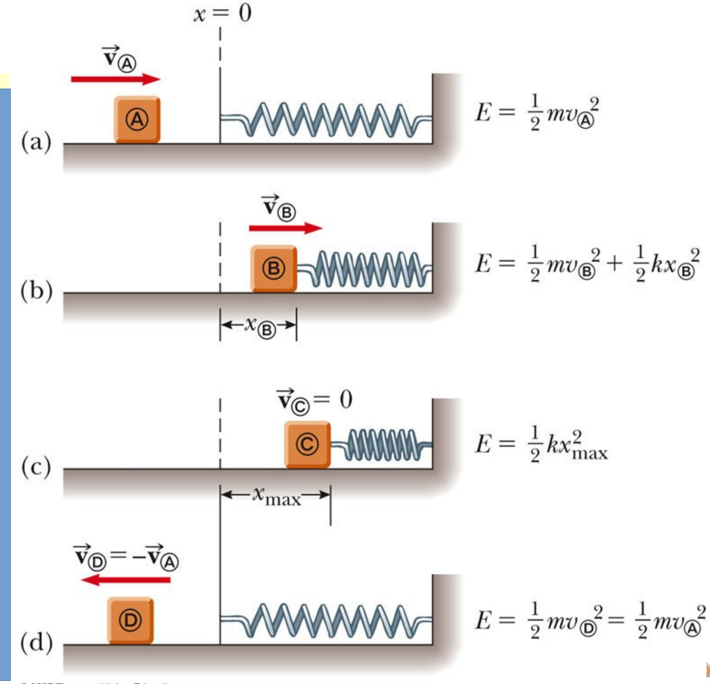
$$-\mu_k mgx=-\frac{1}{2}mv_i^2$$

$$x=\frac{v_i^2}{2\mu_k g}=2.5\,m$$



Block-Spring Collision

- A block having a mass of 0.8 kg is given an initial velocity v_A = 1.2 m/s to the right and collides with a spring whose mass is negligible and whose force constant is k = 50 N/m as shown in figure. Suppose a constant force of kinetic friction acts between the block and the surface, with μ_k = 0.5, what is the maximum compression x_c in the spring.



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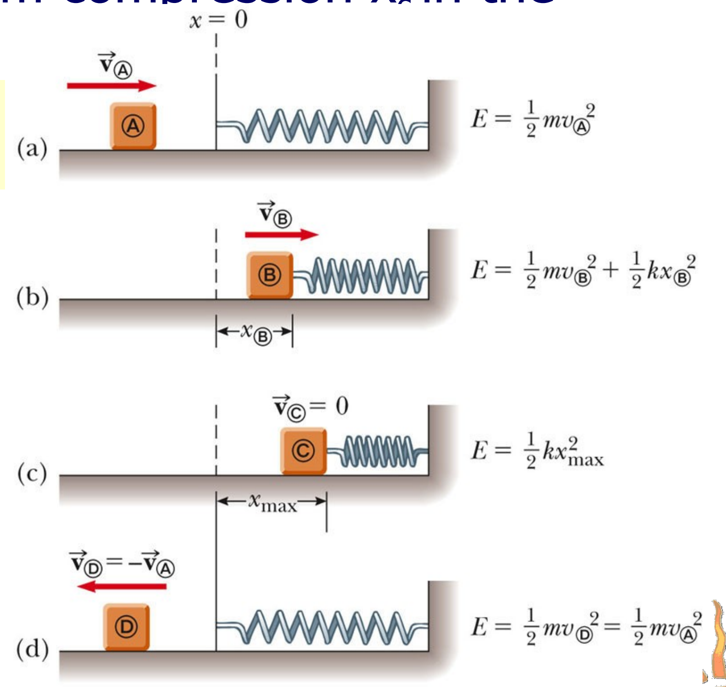
$$-fd+\sum W_{otherforces}=\left(\frac{1}{2}mv_f^2+mgy_f+\frac{1}{2}kx_f^2\right)-\left(\frac{1}{2}mv_i^2+mgy_i+\frac{1}{2}kx_i^2\right)$$

$$-\mu_k Nd+0=\left(0+0+\frac{1}{2}kx_c^2\right)-\left(\frac{1}{2}mv_A^2+0+0\right)$$

$$N=mg \qquad \text{and} \qquad d=x_c$$

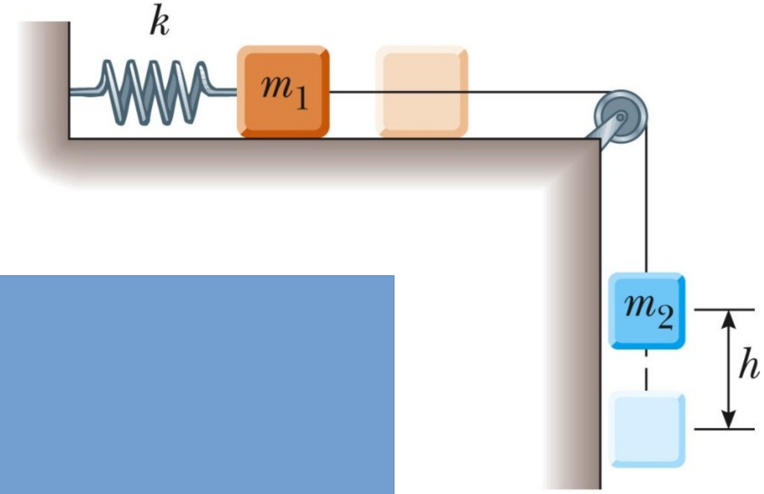
$$\frac{1}{2}kx_c^2-\frac{1}{2}mv_A^2=-\mu_k mgx_c$$

$$25x_c^2+3.9x_c-0.58=0 \qquad x_c=0.093m$$



Connected Blocks in Motion

- Two blocks are connected by a light string that passes over a frictionless pulley. The block of mass m_1 lies on a horizontal surface and is connected to a spring of force constant k. The system is released from rest when the spring is unstretched. If the hanging block of mass m_2 falls a distance h before coming to rest, calculate the coefficient of kinetic friction between the block of mass m_1 and the surface. $-fd+\sum W_{otherforces}=\Delta KE+\Delta PE$



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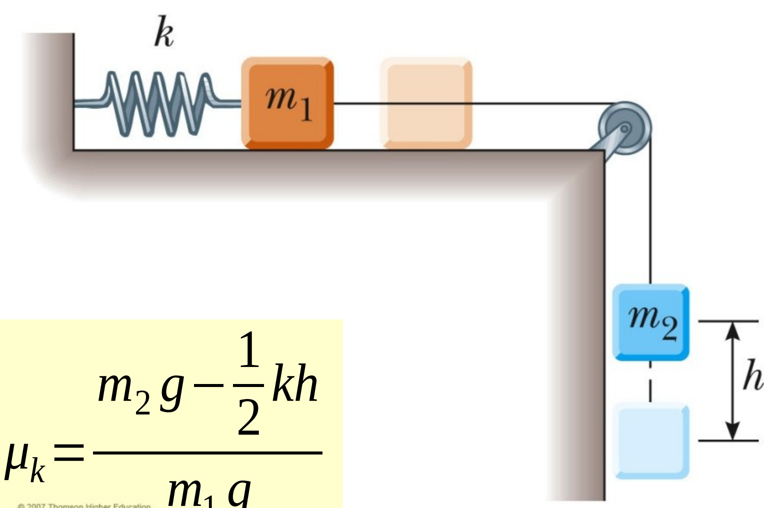
$$\Delta PE=\Delta PE_g+\Delta PE_s=(0-m_2gh)+(\frac{1}{2}kx^2-0)$$

$$-\mu_k Nx+0=-m_2gh+\frac{1}{2}kx^2$$

$$N=mg \quad \text{and} \quad x=h$$

$$-\mu_k m_1gh=-m_2gh+\frac{1}{2}kh^2$$



$$\mu_k=\frac{m_2g-\frac{1}{2}kh}{m_1g}$$

ESCAPE VELOCITY

Fire bullets up and they will come back down ! (some people don't realize that).

You need 7 miles per second to escape Earth gravity. (11km/s).
Interestingly, an object from infinity falls on earth with the same speed.

To find the speed : gravitation energy = kinetic energy.
Solve for the speed.

$$\frac{1}{2}mv_i^2 - \frac{GM_E m}{R_E} = \frac{1}{2}m0^2 - \frac{GM_E m}{\infty}$$

$$\frac{1}{2}mv_i^2 - \frac{GM_E m}{R_E} = 0 - 0$$

$$\frac{1}{2}mv_i^2 - \frac{GM_E m}{R_E} = 0$$

$$V_e = \sqrt{\frac{2GM}{R}}$$

$$= \sqrt{\frac{2 \times 6.6726 \times 10^{-11} \times 5.9742 \times 10^{24}}{6378100}}$$

$$= \sqrt{\frac{2 \times 6.6726 \times 5.9742 \times 10^{24-11}}{6378100}}$$

$$= \sqrt{\frac{79.73 \times 10^{13}}{6378100}}$$

$$= \sqrt{125005879.5} = 11180.6 \text{ m/s}$$

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